Project: Election System (OOT Project)

Modules Covered:

- Main Menu
- Admin Login
- Debug Mode

□ 1. EQUIVALENCE PARTITIONING (EP)

1.1 Main Menu

Test Case ID	Input	Partition Type	Expected Output
MM_EP_01	1–12 (valid options)	Valid	Navigate to selected module
MM_EP_02	0	Invalid	Show error message
MM_EP_03	13+	Invalid	Show error message
MM_EP_04	String ("abc")	Invalid	Show error message

```
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         ELECTION COMMISSION OOT SYSTEM - MAIN MENU
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1.   Voter Portal                2.   Candidate Portal
3.   Admin Portal                4.   Officer Portal
5.   EVM Management              6.   Polling Station Portal
7.   Complaint Portal            8.   Nomination Portal
9.   Constituency Portal         10.  Political Party Portal
11.  Election Portal             12.  Exit

==============================================================

Enter your choice: abc

❌ Invalid Input! Only numbers allowed.
```

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==============================================================
         ELECTION COMMISSION OOT SYSTEM - MAIN MENU
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1.   Voter Portal                2.   Candidate Portal
3.   Admin Portal                4.   Officer Portal
5.   EVM Management              6.   Polling Station Portal
7.   Complaint Portal            8.   Nomination Portal
9.   Constituency Portal         10.  Political Party Portal
11.  Election Portal             12.  Exit

==============================================================

Enter your choice: 45

❌ Invalid Choice! Allowed range is 1 to 12.
```

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         ELECTION COMMISSION OOT SYSTEM - MAIN MENU
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1.   Voter Portal                2.   Candidate Portal
3.   Admin Portal                4.   Officer Portal
5.   EVM Management              6.   Polling Station Portal
7.   Complaint Portal            8.   Nomination Portal
9.   Constituency Portal         10.  Political Party Portal
11.  Election Portal             12.  Exit

==============================================================

Enter your choice: 4

--- OFFICER MENU ---
1. Display
2. Assign Polling Station
3. Monitor EVM
4. Handle Complaint
5. Manage Election
6. Back
```

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         ELECTION COMMISSION OOT SYSTEM - MAIN MENU
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1.   Voter Portal                2.   Candidate Portal
3.   Admin Portal                4.   Officer Portal
5.   EVM Management              6.   Polling Station Portal
7.   Complaint Portal            8.   Nomination Portal
9.   Constituency Portal         10.  Political Party Portal
11.  Election Portal             12.  Exit

==============================================================

Enter your choice: 0

❌ Invalid Choice! Allowed range is 1 to 12.
```

1.2 Admin Login

Test Case ID	Input	Partition Type	Expected Output
AL_EP_01	Correct username & password	Valid	Login successful
AL_EP_02	Wrong username	Invalid	Login failed
AL_EP_03	Wrong password	Invalid	Login failed
AL_EP_04	Empty fields	Invalid	Prompt required fields
AL_EP_05	Special characters in username	Invalid	Error message

```
parth-acharya@parth-acharya-Dell-G15-5530:~/Documents/OOT /Prototype$ ./run
_____________________________________________________________________________________________

------------------- WELCOME TO ELECTION COMMISSION OOT PROTOTYPE -------------------------
_____________________________________________________________________________________________


================ ADMIN LOGIN ================
Enter Admin Name: admin1
Enter Password:

Access Denied! Invalid Credentials.
Attempts left: 2

================ ADMIN LOGIN ================
Enter Admin Name: parth
Enter Password:

Access Denied! Invalid Credentials.
Attempts left: 1

================ ADMIN LOGIN ================
Enter Admin Name: PARTF
Enter Password:

Access Denied! Invalid Credentials.
Attempts left: 0

Too many failed attempts. Exiting system.
parth-acharya@parth-acharya-Dell-G15-5530:~/Documents/OOT /Prototype$
```

```
parth-acharya@parth-acharya-Dell-G15-5530:~/Documents/OOT /Prototype$ ./run
_____________________________________________________________________________________________

------------------- WELCOME TO ELECTION COMMISSION OOT PROTOTYPE -------------------------
_____________________________________________________________________________________________


================ ADMIN LOGIN ================
Enter Admin Name: parth
Enter Password:

Access Granted! Welcome parth 👋

Currently Logged In Admin: parth

Enable Debug Mode? (1 = Yes, 0 = No):
```

```
parth-acharya@parth-acharya-Dell-G15-5530:~/Documents/OOT /Prototype$ ./run
_____________________________________________________________________________________________

------------------- WELCOME TO ELECTION COMMISSION OOT PROTOTYPE -------------------------
_____________________________________________________________________________________________


================ ADMIN LOGIN ================
Enter Admin Name: prajyesh
Enter Password:

Access Granted! Welcome prajyesh 👋

Currently Logged In Admin: prajyesh

Enable Debug Mode? (1 = Yes, 0 = No):
```

1.3 Debug Mode

Test Case ID	Input	Partition Type	Expected Output
DM_EP_01	Valid debug command	Valid	Execute debug
DM_EP_02	Invalid command	Invalid	Show error
DM_EP_03	Empty input	Invalid	Prompt input
DM_EP_04	Numeric input instead of command	Invalid	Error

```
Currently Logged In Admin: parth

Enable Debug Mode? (1 = Yes, 0 = No): 0
[NORMAL MODE]
```

```
Access Granted! Welcome prajyesh 👋

Currently Logged In Admin: prajyesh

Enable Debug Mode? (1 = Yes, 0 = No): 1
[DEBUG MODE ENABLED]
Voter object destroyed: Karan
User object destroyed: Karan
Voter object destroyed: Neha
User object destroyed: Neha
```

```
Enter Admin Name: parth
Enter Password:

Access Granted! Welcome parth 👋

Currently Logged In Admin: parth

Enable Debug Mode? (1 = Yes, 0 = No): 5
[NORMAL MODE]

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□ 2. BOUNDARY VALUE ANALYSIS (BVA)

2.1 Main Menu

Test Case ID	Input	Boundary Type	Expected Output
MM_BVA_01	1	Lower boundary	Valid selection
MM_BVA_02	12	Upper boundary	Valid selection
MM_BVA_03	0	Just below lower	Error
MM_BVA_04	13	Just above upper	Error
MM_BVA_05	2–11	Normal	Valid

2.2 Admin Login (Password length example: 6–12)

Test Case ID	Input	Boundary Type	Expected Output
AL_BVA_01	6 chars	Min valid	Accept
AL_BVA_02	12 chars	Max valid	Accept
AL_BVA_03	5 chars	Below min	Reject
AL_BVA_04	13 chars	Above max	Reject

2.3 Debug Mode

Test Case ID	Input Length	Boundary Type	Expected Output
DM_BVA_01	1 char	Min	Accept/Process
DM_BVA_02	Max allowed	Max	Accept
DM_BVA_03	Empty	Below min	Error

3. DECISION TABLE TESTING

3.1 Admin Login

Conditions	Case 1	Case 2	Case 3	Case 4
Correct Username	Y	Y	N	N
Correct Password	Y	N	Y	N

Action	Login Success	Error	Error	Error

3.2 Main Menu

Conditions	Case 1	Case 2	Case 3
Input valid (1–12)	Y	N	N
Input numeric	Y	Y	N
Action	Navigate	Error	Error

3.3 Debug Mode

Conditions	Case 1	Case 2	Case 3
Valid command	Y	N	N
Input empty	N	Y	N
Input invalid	N	N	Y
Action	Execute	Prompt	Error

□ 4. ERROR GUESSING

(Based on experience & common bugs)

4.1 Main Menu

Test Case ID	Scenario	Expected Output
MM_EG_01	Infinite loop on wrong input	Program should not hang
MM_EG_02	Enter float (2.5)	Reject input
MM_EG_03	Press Enter only	Ask for input again

4.2 Admin Login

Test Case ID	Scenario	Expected Output
AL_EG_01	SQL injection-like input	Reject input
AL_EG_02	Very long input	Handle safely
AL_EG_03	Case sensitivity issue	Validate properly

4.3 Debug Mode

Test Case ID	Scenario	Expected Output
DM_EG_01	Unknown command	Error
DM_EG_02	Repeated commands	Stable execution

DM_EG_03	Invalid sequence	Proper handling